Mrityunjoy Saha

Director's Fellow

Indian Institute of Technology Jodhpur, Rajasthan, India

CFD | Mathematical Modeling | Numerical Simulations

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SUMMARY

- Applied mathematics and CFD researcher with experience in numerical modeling and stability analysis of complex flow systems. Developed computational methods using spectral techniques and scientific computing for fluid dynamics problems. Skilled in C/C++, Python, MATLAB, and HPC-based computations. Interested in applying mathematical modeling and simulation approaches to real-world engineering problems, including aerospace and autonomous systems.

EDUCATION

- **Ph.D. in Fluid Mechanics** **2021–2025**
  Indian Institute of Technology Jodhpur, India
  Thesis Submitted
  CGPA: 8.69/10
- **M.Sc. in Mathematics (Applied Mathematics)** **2018-2020**
  Cooch Behar Panchanan Barma University, West Bengal, India
  CGPA: 8.19/10
- **B.Sc. (Hons.) in Mathematics** **2015-2018**
  Dinhata College, Cooch Behar Panchanan Barma University, West Bengal, India
  Percentage: 62.5

TECHNICAL SKILLS

- **Programming Languages**
  C/C++, Python
- **Scientific Computing & Simulation**
  MATLAB, Mathematica, COMSOL Multiphysics, High Performance Computing (HPC)
- **Domain Expertise**
  Computational Fluid Dynamics (CFD), Hydrodynamic Stability, Fluid Flow Modeling, Numerical PDEs

PUBLICATIONS

- **M. Saha** and S. Ghosh, "On the magnetohydrodynamic instability pattern of an anisotropic porous media flow in a rotating channel," Physics of Fluids, vol. 38(1), 014122, 2026.
- M. M. Hossain, **M. Saha**, S. Ghosh, and H. Behera, "Stability dynamics of an odd viscosity induced fluid flow over a vibrating inclined surface," Physics of Fluids, vol. 37(10), 104108, 2025.
- **M. Saha**, S. Sengupta, S. Mukhopadhyay, and S. Ghosh, "Flow instability in a rotating channel loaded with an anisotropic porous material," Computers & Fluids, vol. 299, 106689, 2025.

RESEARCH IN PROGRESS

- **M. Saha**, S. Ghosh, M. Roy, and M. Mishra, "Modal and non-modal stability of Poiseuille flow through anisotropic inhomogeneous porous materials." (Under review)
- **M. Saha**, S. Ghosh, and K. C. Sahoo, "Dual instability in a vertical crossflow-induced oscillating channel flow." (In preparation)
- S. Panda, **M. Saha**, and S. Ghosh, "Effect of uniform vertical crossflow on rotational instability in a channel flow." (In preparation)

EXPERIENCE

- **Teaching Assistant** **2022–Present**
  Indian Institute of Technology Jodhpur
  Courses assisted: Mathematics I & II, Scientific Computation, Mathematical Modeling, Data Science, Applied Mathematics & Statistics, Linear Algebra, Complex Analysis.

PROFESSIONAL TRAINING

- **Python Programming for Computational Fluid Dynamics Applications** **June-July, 2024**
  Electronics & ICT Academy, Indian Institute of Technology Guwahati

CONFERENCE PRESENTATIONS

- **78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2025)**
  George R. Brown Convention Center, Houston, Texas, USA.
- **International Conference on Recent Advances in Fluid Mechanics and Nanoelectronics 2023**
  Manipal Institute of Technology, Bengaluru, India.

AWARDS AND SCHOLARSHIPS

- Director's Fellowship, Indian Institute of Technology Jodhpur.
- American Physical Society - Division of Fluid Dynamics Travel Award 2025.
- ANRF International Travel Support for APS DFD 2025.
- CSIR Foreign Travel Grant for APS DFD 2025.
- Qualified Graduate Aptitude Test in Engineering (GATE), Mathematics (2022, 2023).
- DST INSPIRE Scholarship - Government of India.
- SVMCM Scholarship (UG, PG, and Higher Secondary).
- Sitaram Jindal Scholarship for B.Sc.

REFERENCE

- **Dr. Sukhendu Ghosh** (Ph.D. Supervisor)
  Associate Professor, Department of Mathematics, IIT Jodhpur
  Email: sukhendu.math@iitj.ac.in